

NAPOLEON TORRICO

Los Angeles, CA • (818) 800-2052 • napoleontorricoiv@gmail.com
linkedin.com/in/napoleon-torrico-iv/ • github.com/nappalion • napoleontorricoiv.com

EDUCATION

California State Polytechnic University, Pomona | Pomona, CA

Aug 2021 – Dec 2023

Bachelor of Science in Computer Science, Summa Cum Laude (3.98)

Relevant coursework: Software Engineering, Mobile Application Development, Object-Oriented Design and Programming, Design and Analysis of Algorithms, Big Data Analytics, and Cloud Computing

CodePath | Remote

Certificate in Intermediate Software Engineering

Jun 2022 – Aug 2022

SKILLS

- Languages: TypeScript, JavaScript/HTML/CSS, Python, Java, Kotlin, SQL
- Tools & Frameworks: React Native, Expo, React, Node.js, Express, PostgreSQL, Prisma, SQLite, Zustand, React Query, Jest, Vitest, Docker, Nginx, Tailwind, Passport.js, Firebase, Jetpack Compose, Android Studio, Figma, Git

EXPERIENCE

Senior Computer Science Instructor | Juni Learning

Oct 2021 – Nov 2025

- Curated unique advanced computer science lesson plans in the fundamentals of Java, Python, and OO design over Zoom for students ages 8-18 that adapt to support each student's needs and interests.

Web Development 101 Tech Fellow | CodePath

Jan 2024 – Apr 2024

- Mentored **4 groups of 5 students** each, leading them through a comprehensive curriculum including concepts in HTML, CSS, and JavaScript, resulting in a better understanding of the material and increased participation.
- Provided technical assistance on Slack for the **200-student** course while fostering an inclusive learning environment by establishing professional connections among and with students, nurturing their motivation to learn.

PROJECTS

PomoTomo | Personal Project

- Developed and published a full-stack social productivity app to the App Store and Google Play using TypeScript, React Native, Expo, Node.js, and PostgreSQL, featuring live study channels with real-time synchronized timers, a social feed of focus sessions, user following, direct messaging, and live chat via WebSockets.
- Engineered an **offline-first** architecture with local SQLite storage, idempotent action queuing, server-side integrity validation, and background sync, enabling reliable focus tracking without network connectivity.
- Secured the API with **JWT authentication** and rotating refresh tokens, Google and Apple OAuth 2.0 with server-side ID token verification, per-IP and per-user rate limiting, Zod schema validation, and HTTPS via Nginx.
- Achieved **85%+ code coverage** across **625+ tests** using Jest, React Native Testing Library, and Vitest in a TypeScript monorepo, and automated testing, container builds, and deployment via GitHub Actions and Docker.

Icebreak - Frontend Team Lead | CPP Software Engineering Association

- Led a team of **9 students** using a Node.js API and React Native to create an organization management platform that incentivizes member growth.
- Adhered to **Agile**-like software development lifecycle practices, including RFCs, tests, and code reviews.
- Organized weekly standups and collaborated with team members to complete development sprints.

Pallet Configuration | Personal Project

- Designed a pallet configuration algorithm inspired by Tetris-solving AI in JavaScript, using React Native for the UI and Firebase for storage.
- Achieved an average runtime reduction of **85%** from **6.52 to 0.98 seconds** for **100** different boxes by employing array optimization and caching techniques.
- Increased pallet load efficiency by **15%** with an average of **6.22** more boxes per pallet by comparing algorithms for static and dynamic box rotation.
- Employed the Three.js library to generate and render 3D graphics for pallet layouts dynamically.